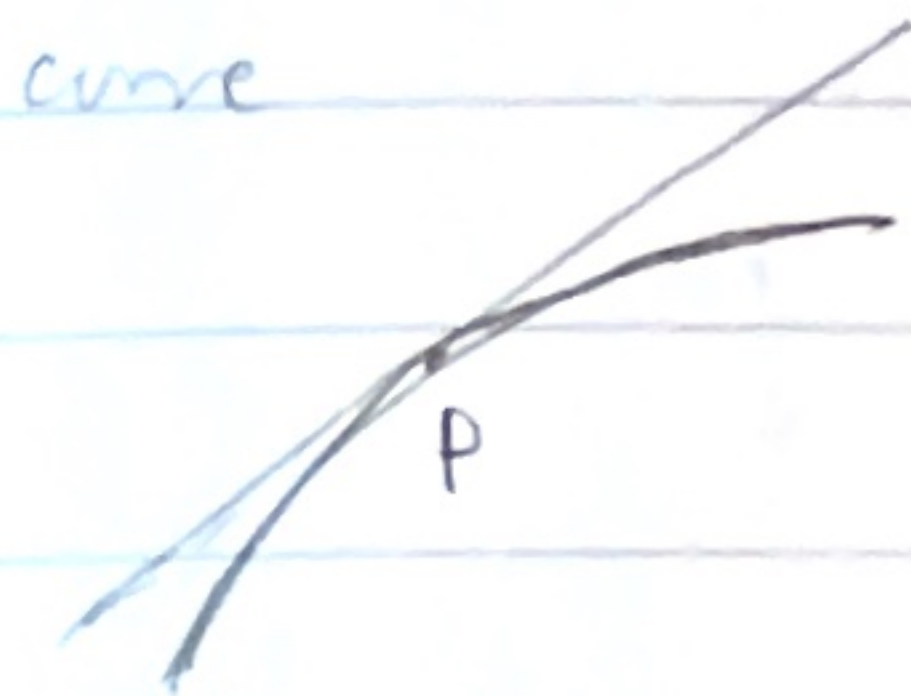
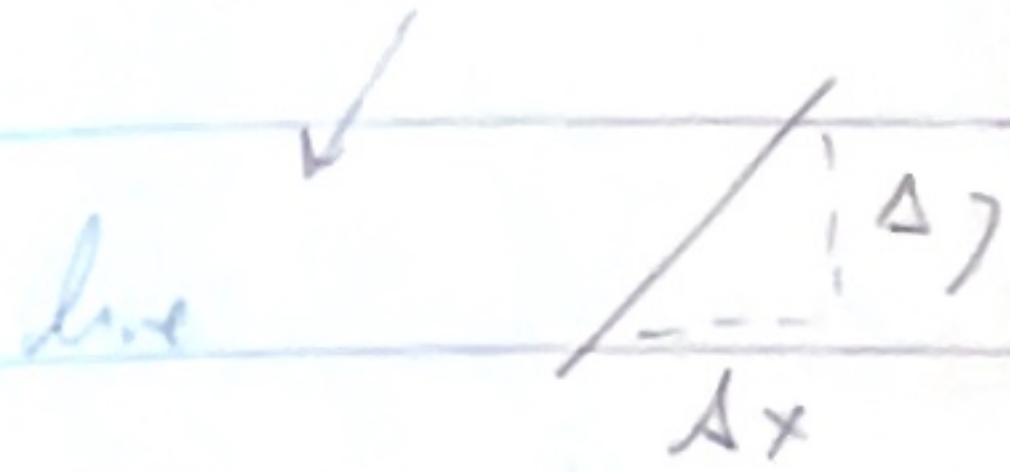


Differentiation

$$\text{Slope} = \frac{\Delta y}{\Delta x}$$



$$\left[\text{Slope at } P: \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \right]$$

$$\text{Speed} \rightarrow \text{constant } \frac{\Delta x}{\Delta t}$$

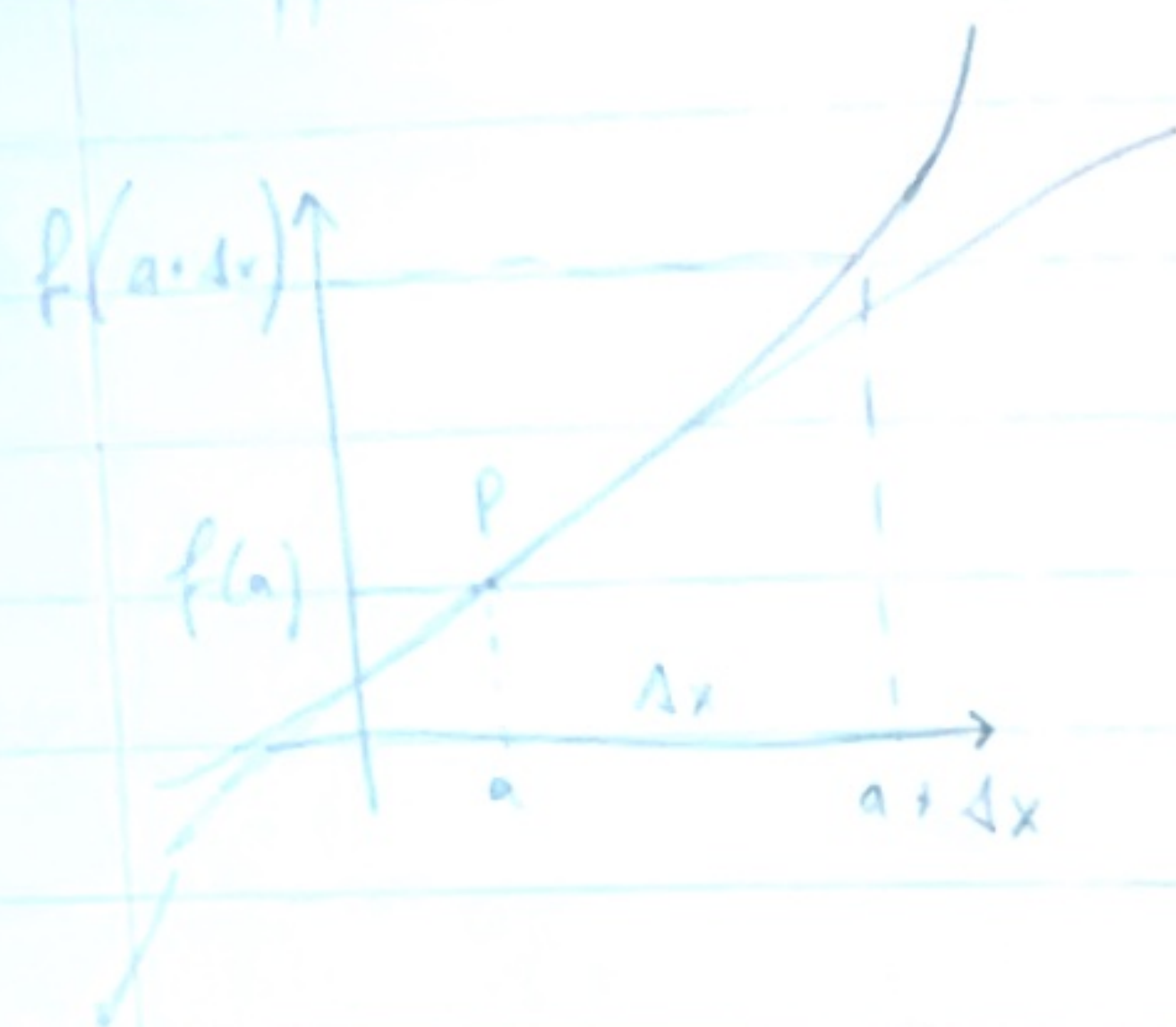
$$\text{no constant } \frac{\Delta x}{\Delta t} = \text{average speed}$$

↓
small $\approx$ speed at time t

$$\left[\lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} \right]$$

defining the derivative

Differentiation - definition



Difference quotient

$$\frac{f(a+\Delta x) - f(a)}{\Delta x}$$

slope of line PA:

$$\frac{f(a+\Delta x) - f(a)}{\Delta x}$$

the limit is

$$\lim_{\Delta x \rightarrow 0} \frac{f(a+\Delta x) - f(a)}{\Delta x}$$

is limit

Tangent line
at P
slope: $f'(a)$

if limit exists the
derivative

Notation

$$f'(a) \text{ or } \frac{df}{dx}(a)$$

Rules of calculation:

1) Standard derivatives

2) rules of calculation

→ Multiplication by a constant

$$[cf(x)]' = cf'(x)$$

→ Addition

$$[f(x) + g(x)]' = f'(x) + g'(x)$$

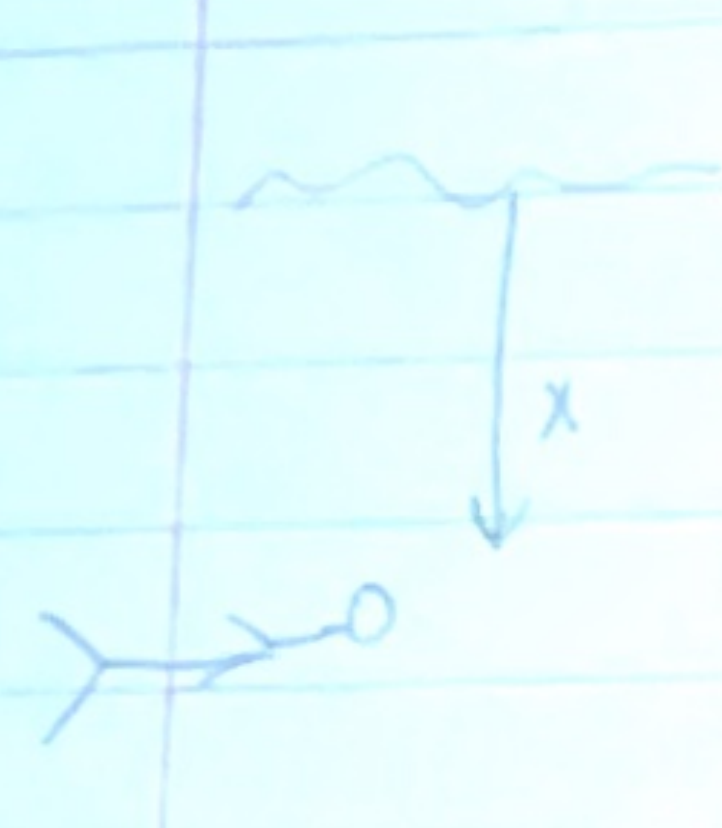
→ Product rule (Leibniz rule)

$$[w(x)l(x)]' =$$

$$w(x)l'(x) + w'(x)l(x)$$

Chain Rule

due in sec $\vec{p} = p(x(t))$



$$\frac{\Delta p}{\Delta t} = \frac{\Delta p}{\Delta x} \cdot \frac{\Delta x}{\Delta t}$$

$$\frac{dp}{dt} = \frac{dp}{dx} \cdot \frac{dx}{dt}$$

$$\downarrow$$

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

Quotient rule =

$$\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{d}{dx} f(x) \cdot \frac{1}{g(x)} \quad \left| \quad \frac{d}{dx} \frac{1}{x} = \frac{d}{dx} x^{-1} = -x^{-2} \right.$$

$$= f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \left(\frac{1}{dx} \frac{1}{g(x)} \right)$$

$$= \frac{f'(x)}{g(x)} + f(x) \cdot \frac{-1}{g(x)^2} \cdot g'(x)$$

$$= \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g(x)^2}$$

$$\frac{d}{dx} \frac{x}{x^2+1} = \frac{1}{x^2+1} + x \cdot \frac{-1}{(x^2+1)^2} \left(\frac{d}{dx} (x^2+1) \right)$$

$$= \frac{x^2+1}{(x^2+1)^2} + \frac{x(-1) \cdot 2x}{(x^2+1)^2}$$

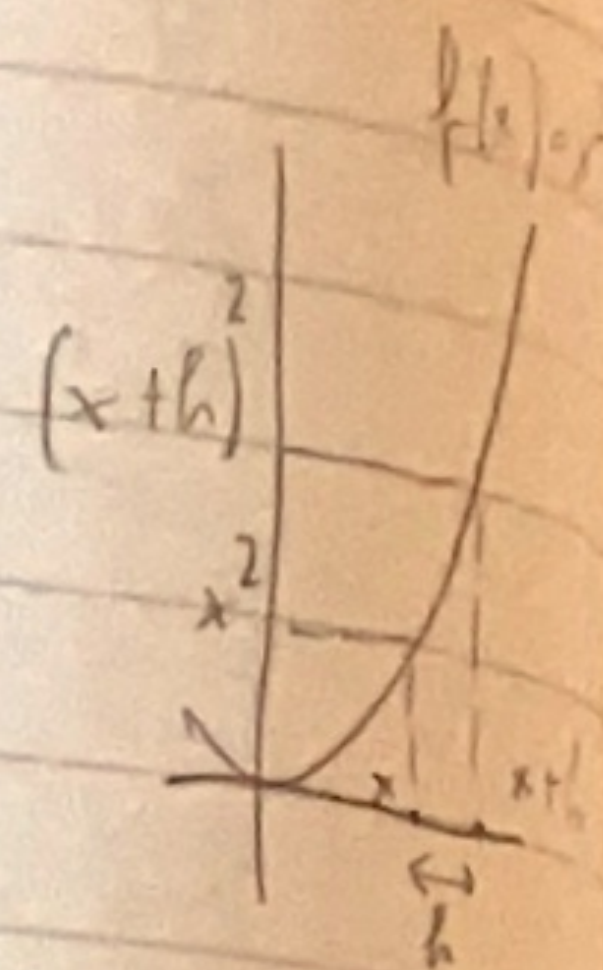
$$= \frac{1-x^2}{(x^2+1)^2}$$

derivatives of power functions

x^n $n \geq 0$, integer

ex $f(x) = x^2$

$$\begin{aligned} \frac{f(x+h) - f(x)}{h} &= \frac{(x+h)^2 - x^2}{h} \\ &= \frac{x^2 + 2hx + h^2 - x^2}{h} \\ &= 2x + h \end{aligned}$$



$f(x)$	derivative
1	0
x	1
x^2	$2x$
x^3	$3x^2$

To calculate the derivative of the cubic $f(x)$

$$f(x) = x^3$$

$$f'(x) = \frac{d}{dx} (x \cdot x^2)$$

$$= \left(\frac{d}{dx} x \right) \cdot x^2 + x \left(\frac{d}{dx} x^2 \right)$$

$$= 1 \cdot x^2 + x \cdot 2x$$

$$= 3x^2$$

CONTANT EXPONENTS

$$f'(x^n) = n \cdot x^{n-1}$$

Negative n

$$\rightarrow \frac{d}{dx} \frac{1}{x} = \frac{d}{dx} x^{-1} = -x^{-2} = -\frac{1}{x^2}$$

Fractional n

$$\frac{d}{dx} \sqrt{x} = \frac{d}{dx} x^{1/2} = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$$

any real number n

$$\frac{d}{dx} x^n = n x^{n-1}$$

POLYNOMIALS

$$\frac{d}{dx} (2 + 4x + 5x^2 + 7x^3 - x^4) =$$

$$2 \cdot 0 + 4 \cdot 1 + 5 \cdot 2x + 7 \cdot 3x^2 - 4x^3$$

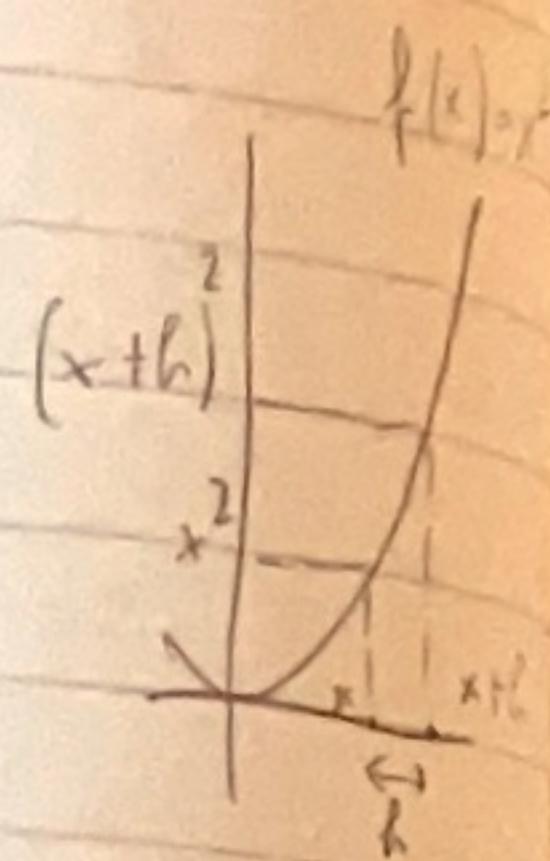
derivatives of power functions
 x^n $n \geq 0$, integer

ex $f(x) = x^2$

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h}$$

$$= \frac{x^2 + 2hx + h^2 - x^2}{h}$$

$$= 2x + h$$



$f(x)$	derivative
1	0
x	1
x^2	$2x$
x^3	$3x^2$

To calculate the derivative of the cubic $f(x)$
 $f(x) = x^3$

$$f'(x) = \frac{d}{dx} (x \cdot x^2)$$

$$= \left(\frac{d}{dx} x \right) \cdot x^2 + x \left(\frac{d}{dx} x^2 \right)$$

$$= 1 \cdot x^2 + x \cdot 2x$$

$$= 3x^2$$

CONTANT EXPONENTS

$$f'(x^n) = n \cdot x^{n-1}$$

Negative n

$$\rightarrow \frac{d}{dx} \frac{1}{x} = \frac{d}{dx} x^{-1} = -x^{-2} = -\frac{1}{x^2}$$

Fractional n

$$\frac{d}{dx} \sqrt{x} = \frac{d}{dx} x^{1/2} = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$$

any real number n

$$\frac{d}{dx} x^n = n x^{n-1}$$

POLYNOMIALS

$$\frac{d}{dx} (2 + 4x + 5x^2 + 7x^3 - x^4) =$$

$$2 \cdot 0 + 4 \cdot 1 + 5 \cdot 2x + 7 \cdot 3x^2 - 4x^3$$

SINE AND COSINE

$\cos \theta$

derivative of sine equals ~~the derivative~~ of
AOR

$$e^x$$

e is the unique number such that

derivative of $e^x = e^x$

Using Chain Rule

$$\begin{aligned}\frac{d}{dx} a^x &= \frac{d}{dx} e^{x \ln(a)} \\ &= e^{x \ln(a)} \cdot \frac{d}{dx} x \ln(a) \\ &= a^x \cdot \ln(a)\end{aligned}$$

$$y = \ln(x) \quad \rightarrow \quad x = e^{\ln(x)}$$

Chain rule to one side

$$1 = e^{\ln(x)} \cdot \frac{d}{dx} \ln(x)$$

$$1 = x \cdot \frac{d}{dx} \ln(x)$$

$$\frac{d}{dx} \ln(x) = \frac{1}{x}$$

TANGENT LINE

Slope $f'(p)$
through $(p, f(p))$

Equation tangent line

$$y = f'(p)(x-p) + f(p)$$

non-differentiable functions

the function f is differentiable at a if
 $\lim_{\Delta x \rightarrow 0} \frac{f(a + \Delta x) - f(a)}{\Delta x}$

exists.

by "exists" we mean the limit must be exactly one real number

examples:

- a discontinuous function
- graph with a kink
- tangent line is vertical $\rightarrow$ slope is ∞

- more "exotic" functions such as $x \sin\left(\frac{1}{x}\right)$

finding local minima and maxima

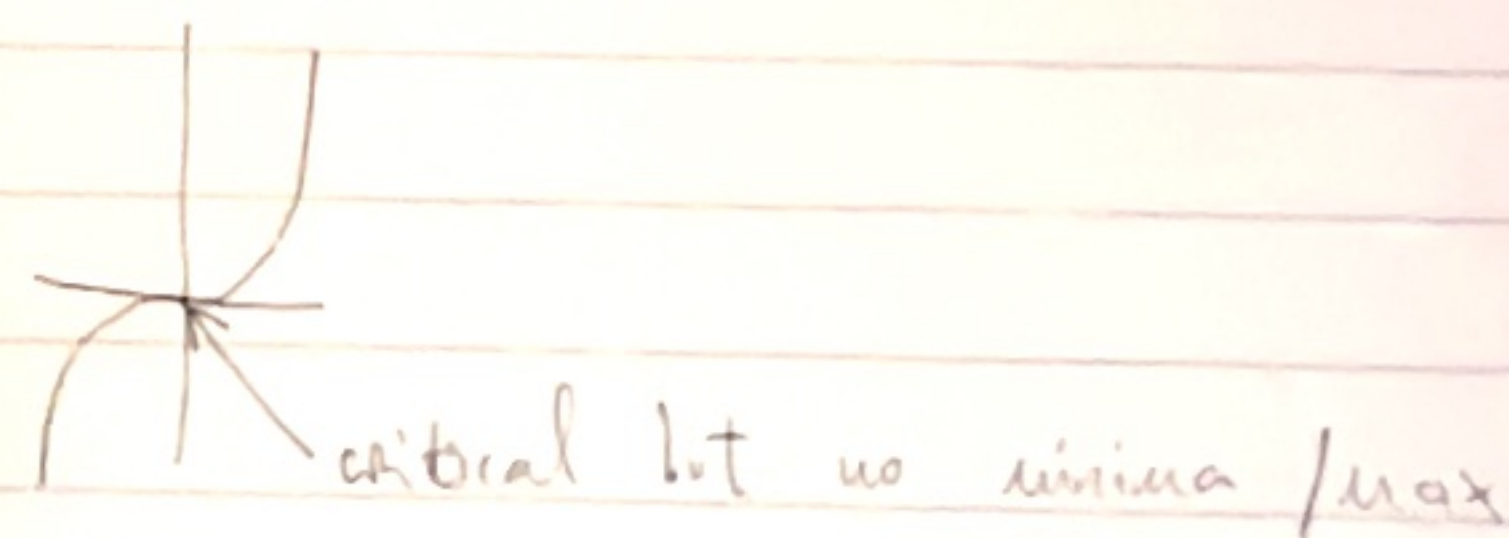
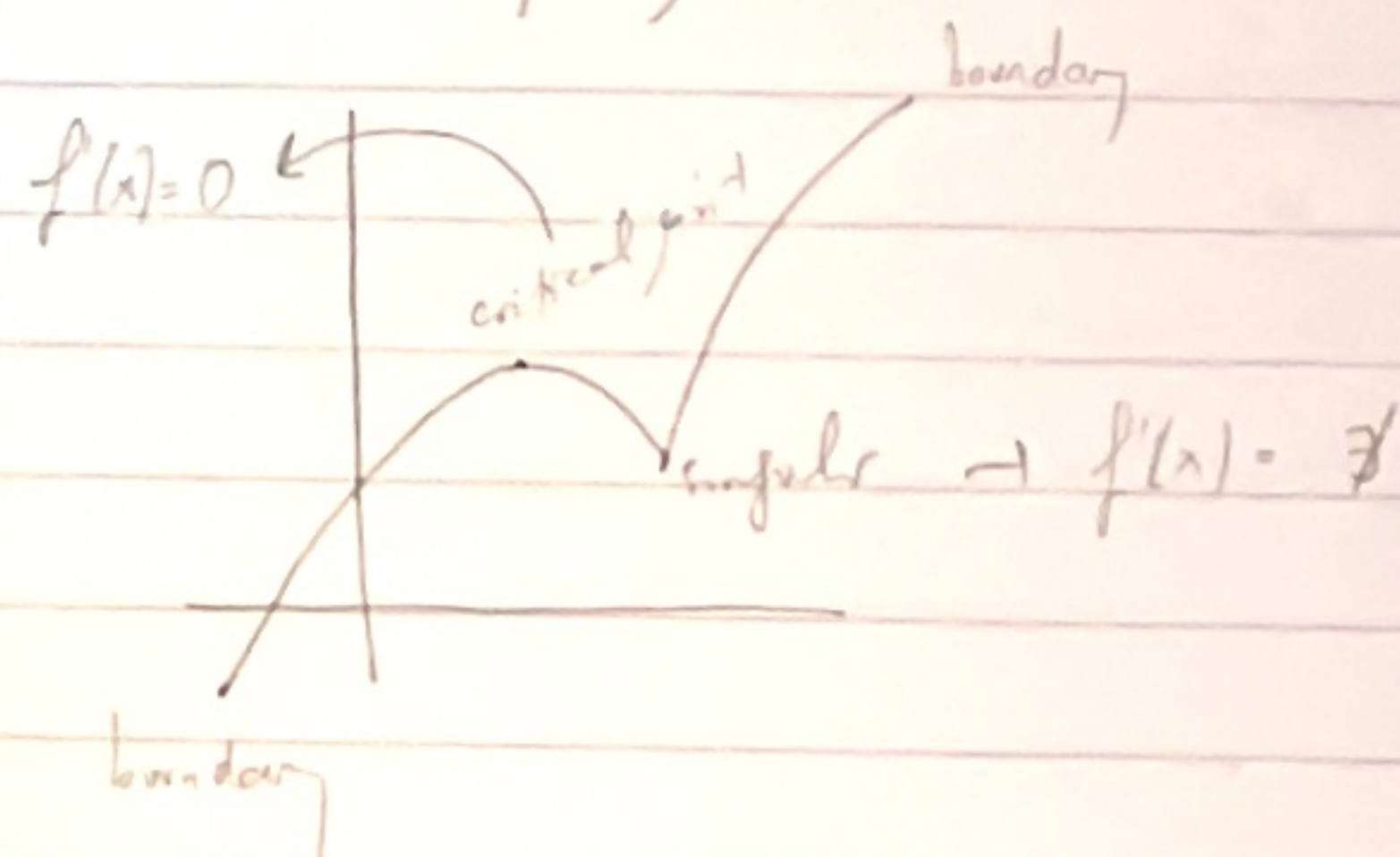
optimization

$$f'(x) > 0 \Rightarrow \text{increase}$$

$$f'(x) < 0 \Rightarrow \text{decrease}$$

$$\text{at extrema } f'(x) = 0$$

!!
① non-differentiable functions
boundary points



1) find boundary, critical and singular points